The Distance-to-Mean Broadcast Method for Vehicular Wireless Communication Systems

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Abstract— **Broadcast is a critical component in embedded communications systems. Some vehicular network (VANET) applications in particular use broadcast communications extensively. Statistical broadcast methods offer an efficient means of propagating broadcast messages in this context due to their low overhead and high efficiency.**

Currently, four fundamental statistical broadcast methods are known: stochastic, counter, distance, and location. In this work, we introduce a new method called distance-to-mean. Utilizing positional information, this method calculates the spatial mean of the neighbors from which a node has received the message, then finds the distance from the node to that spatial mean. This distance is used as the variable to discriminate between rebroadcasting and non-rebroadcasting nodes.

Simulation results comparing distance, location, and distance-to-mean show that distance-to-mean exhibits high transmit efficiency while maintaining similar network coverage. The location method is more complex to calculate than the distance-to-mean method, so these results suggest that the distance-to-mean method should replace the location method in applications where positional information is available.

I. Introduction

Wireless broadcast plays an important role in the operation of embedded wireless systems. Research in this area has demonstrated that blindly retransmitting broadcast packets (flooding) can lead to an explosive growth of traffic called the *broadcast storm* problem [1], that reduces battery life and attenuates communication performance as a result of collisions and congestion.

Further, emerging classes of embedded systems use broadcast as a primary data delivery mechanism. Vehicular Ad-hoc NETworks (VANETs) are a class of wireless networks in which mobile vehicles communicate with each other in an ad-hoc fashion. VANET applications such as traffic data dissemination use broadcast exclusively as the means of communication, so solving the broadcast storm problem in an efficient way is critical.

Dedicated Short Range Communications (DSRC), a layer 1 and 2 standard based on IEEE 802.11 designed for vehicle-to-vehicle and vehicle-to-infrastructure communications, allows transmission ranges up to 1000m. Network density in VANET therefore can range from one or two neighbors to very high densities. Even with a more modest transmission radius of 200m, the number of vehicles in range of any point on a 4-lane road with cars separated by 20m is 80. Broadcast protocols should be able to function well across this broad range of densities.

When network density is high, the communication system faces a particularly adverse environment during a broadcast. If all nodes rebroadcast source messages, then network power consumption will be very high, making it difficult for power-constrained systems to support. Even when power is not a concern, as in VANET, the MAC protocol will be severely hampered by contention on the medium. Collisions in this situation can cause the broadcast propagation to prematurely die out.

Communication algorithms for solving this problem can be classified into at least statistical and topological methods [2]. Topological protocols can be proven to provide full coverage as long as the network is connected and no one-hop link errors occur. Often these protocols use the network topology (knowledge of the identity of nodes in the 2-hop neighborhood) to accomplish this.

In the mobile embedded environment, these topological protocols have several disadvantages. The density and speed of nodes in these systems means that the network topology will be highly variable, so periodic message exchanges will need to be frequent in order to maintain topological information. This will reduce battery life and further aggravate contention and collisions.

Statistical broadcast protocols do not suffer this disadvantage. Statistical protocols measure the value of one or more locally available variables and make a decision to rebroadcast based on the measured value and a cutoff threshold. For example, a statistical protocol used in this work, the distance method, measures the distance to the nearest neighbor from which a node has received the broadcast message. If that distance is greater than a threshold value, then the node rebroadcasts the message. Unfortunately, statistical protocols are stochastic in nature and usually cannot be proven to always give full reachability. Instead, simulation modeling is done to verify that the protocols exhibit acceptable performance in an acceptable percentage of randomly distributed networks.

Currently, there are four published fundamental statistical broadcasting methods: stochastic, counter, distance, location. Each of these methods uses a different variable to discrim-

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inate between broadcasting and non-rebroadcasting nodes. Stochastic broadcast uses a random number generated within each node, counter uses message count, distance uses the distance to the nearest neighbor from whom the message has been received, and location uses the amount of a node's transmission area that has been covered by other neighbors' transmissions. In this work, we present a new fundamental statistical method called distance-to-mean.

In Section I-A we will discuss related work to describe where this work fits into the literature and Section II gives some wireless broadcast preliminaries. Section III presents statistical broadcasting methods, with some existing methods described in Section III-A and the new proposed method in Section III-B. Finally Section IV presents results comparing the new method with the existing methods.

A. Related Work

The fundamental statistical methods (stochastic, counter, distance, and location), are described in [1]. Several protocols have been built using these methods that are adaptive to node density. Several papers have addressed the need to make stochastic broadcast adaptive to density [3] [4] [5]. Density-adaptive versions of counter are given in [6] [7]. Density-adaptive versions of both counter and location methods in [8]. We propose a version of the distance method that is adaptive to both density and link errors in [2]. This work proposes a new statistical method, distance-to-mean, and gives a density-adaptive version of it.

II. Preliminaries

The wireless broadcast problem involves a number of wireless nodes distributed in a field, one of which has a message to be distributed to the rest of them. A broadcast protocol is the mechanism for propagating that message through the field. The simplest broadcast protocol, when nodes always rebroadcast a message the first time they receive it, is called flooding.

Here, nodes are distributed in a field uniformly and each node has a transmission radius r. Throughout this paper, we define density (λ) as the average number of neighbors of each node in the network. The field of continuum percolation informs us that when simple flooding is used in a two-dimensional uniformly distributed field, the network exhibits a behavioral phase transition at a specific value of λ called λ_c. When $\lambda < \lambda_c$, the network is unlikely to be connected. The exact value of λ_c is not known, but it has been estimated to be about 4.5 [9]. In practice, we have found that when $\lambda > 6$, the probability the network is fully connected is greater than 95% [10].

The primary constraint on broadcast protocols is an application specific reachability target. Reachability can be quantified several ways, but in this and most other papers it is defined as the fraction of nodes connected to the source that receive the broadcast message. In practical terms, reachability is more of a functional quality than a performance quality. Performance metrics of protocols should only be compared under conditions where the protocols exhibit similar reachability.

Once protocols are shown to have similar reachability, the first performance metric usually examined is bandwidth utilization. In broadcast, this is some measure of the number of transmissions generated during the propagation of a message. There is some variability in the literature on the best metric for bandwidth utilization. We have presented some disadvantages of using existing metrics such as total number of transmissions or Saved ReBroadcasts (SRB) and introduced a new metric called transmit density [11], which we use here. Transmit density is defined as

$$\lambda_{tx} = \frac{N_{tx}}{A_{field}} \pi r^2 \tag{1}$$

where N_{tx} is the number of nodes that transmit, A_{field} is the size of the field, and r is the transmission radius. More intuitively, λ_{tx} can be viewed as the average number of neighbors that transmit.

III. Statistical Broadcast Protocol

Wireless broadcasting methods can be divided into at least statistical and topological methods [11]. Statistical methods uses a priori knowledge of the impact of an external variable present at each node to decide whether or not to rebroadcast. These methods share a common framework where nodes measure the value of a variable at its location, calculate a threshold value, and make a rebroadcast determination based on whether or not the value of the variable exceeds the value of the threshold. The threshold value is the key to the efficiency and effectiveness of the protocol; if the threshold is set too aggressively, the protocol will exhibit low reachability, too conservative and the broadcast propagation will be inefficient. We denote these methods *statistical* because the threshold function represents an aggregation of knowledge about the statistics of the underlying variable and how the network will perform when retransmitting nodes are selected by partitioning the nodes using the value of that variable. For non-trivial variables and node distributions, the optimal threshold function cannot be determined analytically. Instead, high-level simulation is used to empirically discover good threshold curves.

Statistical methods all share a similar rebroadcast decision algorithm. First, when a node receives a message, it begins measuring the value of some external variable. This measurement may be immediate, or may involve using a backoff timer to assess if more copies of the message will be received. Once the variable has been measured, the node then calculates the value of a threshold function and rebroadcasts if the measured value exceeds the threshold (depending on the nature of the variable, "exceeds" may imply the value is less than or greater than the threshold). In this paper, all the methods utilize a backoff timer, so the algorithms can be summarized as follows, where T is the external variable being measured. Without loss of generality, we assume that the nature of T is such that rebroadcasting nodes have a value of T that is greater than the threshold.

1) Initialize T, calculate threshold value T_c.

2) When a message is received, update the value of T. If $T < T_c$, do not rebroadcast and stop procedure. Otherwise, set a random backoff timer
3) If a message is received during the backoff, repeat 2
4) When the backoff expires, rebroadcast if $T > T_c$

A. Existing Statistical Methods

For comparison purposes, we will look at two existing statistical broadcasting methods: distance and location. The distance method uses the distance to the nearest neighbor from whom the node has received the message as the discriminatory variable. The method appeals to the intuition that if a node has received a message from another node very close to it, there is little benefit in terms of additional coverage achieved by rebroadcasting. Nodes then should favor rebroadcasting when this distance D (which has been normalized to a value between 0 and 1 by dividing by the transmission radius r), is large.

The key to any statistical broadcast protocol is the threshold function, in this case D_c. If the value is set too high, reachability will be degraded. If the value is set too low, the protocol will not prevent many nodes from rebroadcasting. Like the node density threshold λ_c between connected and unconnected networks (see Section II), the value of D_c gives a sharp transition between a network that has high reachability and one in which the message does not propagate [11]. The goal is to set D_c as high as possible without falling off the edge.

Further, this optimal value for D_c varies with node density. At low densities, D_c must be lower to allow the message to propagate to the entire network. When the node density is high, we can set D_c more aggressively to eliminate excess transmissions. Therefore, we set D_c as a function of N, the number of neighbors of the node (also referred to as local density).

Through experimentation, we have found that a good threshold value in terms of the local density is given by Equation 2 [2].

$$D_c(N) = 0.86 - 0.95e^{-0.13N} \tag{2}$$

The location method is similar to distance, except it uses positional information, such as that from GPS, to more accurately measure the amount of a node's transmission area that has already been covered by rebroadcasting neighbors. It calculates the fraction of its transmission area that has not been covered by the received messages and favors rebroadcasting when that fraction is high.

The minimum intersection between two circles of the same radius that contains the centers of both circles has an area about 39% of the circle's area. To normalize the uncovered fraction so it is in the range 0 to 1, we divide the value by 0.61. Specifically, if c is the fraction of the node's transmission area that has been covered by other neighbors rebroadcasting, then the location variable, L, is defined as $L = \frac{1-c}{0.61}$ so it is always in the range from 0 to 1. Through experimentation, we have found that a good threshold function for the location variable is given by Equation 3 [11].

$$L_c(N) = 0.92 - 1.0e^{-0.11N} \tag{3}$$

The location has the disadvantage that it is relatively difficult to calculate. Calculating the fraction of the transmission area covered by transmitting neighbors amounts to finding the union of the intersections of several pairs of circles, for which in general there is no closed form. Here, we use a relatively computationally expensive grid-filling estimation method to approximate this area.

B. Proposed Distance-to-Mean Statistical Method

In this section we propose a new statistical method called distance-to-mean. Like distance and location, distance-to-mean uses the heuristic argument that nodes should favor rebroadcasting when relatively little of their transmission area has already been covered by other nodes. The distance method approximates this by using the distance to the nearest neighbor from whom the broadcast message has been received as a proxy for coverage, and the location method uses positional information to estimate the coverage area directly.

The distance-to-mean variable estimates coverage by considering the distance from the node to spatial mean (defined below) of the neighbors from whom the message has been received already. If this distance is small, it means transmitting neighbors are evenly distributed around the node so it should refrain from retransmitting the message yet again. Like location, the distance-to-mean method requires positional information, but unlike location is very easy to calculate. Calculating the spatial mean of a set of points (x_i, y_i) is simple:

$$(\overline{x}, \overline{y}) = \left(\frac{1}{n} \sum_{i=1}^{n} x_i, \frac{1}{n} \sum_{i=1}^{n} y_i \right) \tag{4}$$

The distance-to-mean variable, M, is the distance from the node to the spatial mean of the nodes it received the message from, normalized to a value between zero and one by dividing by the transmission radius r. If the node is positioned at (x, y), then M is calculated as in Equation 5.

$$M = \frac{1}{r}\sqrt{(x-\overline{x})^2 + (y-\overline{y})^2} \tag{5}$$

Like any statistical method, distance-to-mean uses a threshold value to discriminate between nodes that rebroadcast and those that do not, which is a function of the neighbor count N. Through experimentation, we have found that a good threshold function for the distance-to-mean variable is given by Equation 6.

$$M_c(N) = 0.95 - 0.74e^{-0.11N} \tag{6}$$

IV. Simulation Results

The results presented throughout are found using the Wireless Broadcast Design and Analysis Tool (WiBDAT) [12], a simulation tool developed in Java specifically for evaluating the efficiency of wireless broadcast protocols. Layer 1 (fading)

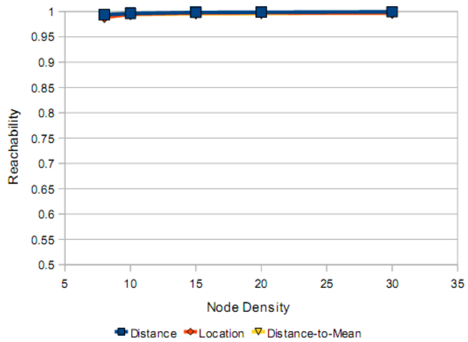

Fig. 1. Reachability vs node density of the statistical methods

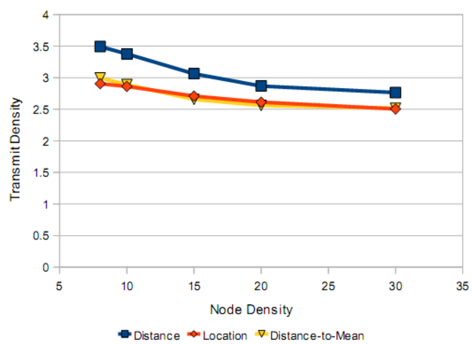

Fig. 2. Transmit density vs node density of the statistical methods

and layer 2 (collisions) effects are ignored, so that the results shown here are purely derived from the behavior of the broadcast protocols under evaluation.

The simulation first distributes a number of homogeneous nodes with transmit radius 10 in a field sized 400 x 400. The node closest to the center of the field is chosen to originate a single broadcast. When the broadcast dies out, the simulation run is complete. Each data point presented is an average of 30 broadcasts.

Two sets of results are presented for each protocol: reachability and transmit density. Reachability is defined here as the fraction of nodes connected to the source that receive the message. For the statistical protocols shown here, threshold values are set to give an efficient trade-off between reachability and transmit density. In this paper, we chose 95% as the minimum acceptable reachability. The second set of results presented is the protocol transmit density, defined by Equation 1

Figures 1 and 2 show the comparison. The first graph, reachability, shows that each protocol fulfills the requirement of adequate broadcast coverage. This comparison is necessary to ensure broadcast efficiency is evaluated under conditions of equivalent functionality.

The second graph shows the transmit density comparison. Location and distance-to-mean are clearly more efficient than distance. The performance of location and distance-to-mean are similar. Since distance-to-mean is a simpler method than location, clearly distance-to-mean is the statistical method of choice when positional information is available.

V. Conclusion

We have presented a new fundamental statistical wireless broadcast communication method called distance-to-mean. This method first uses a random backoff period to collect copies of a broadcast message from transmitting neighbors. Utilizing positional information, it calculates the spatial mean of the neighbors from which a node has received the message, then finds the distance from the node to that spatial mean. When this distance is small, the node has either received the message from nodes that are close by or from nodes that are spaced evenly around it. The distance is large when the sending nodes are clustered away from the receiving node. Nodes favor rebroadcasting when the distance is large.

Simulation results are then presented showing that distance-to-mean out-performs the distance method, and often performs better than the location method. Given that the location method and the distance-to-mean method have similar requirements (both need positional information) and their performance is similar, the introduction of the distance-to-mean method makes the location method obsolete in many cases, since distance-to-mean is much easier to calculate than location.

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